

Practical No. 4

Title : Color Image Segmentation algorithm development

Aim : Color-Based Segmentation using K-Means Clustering.

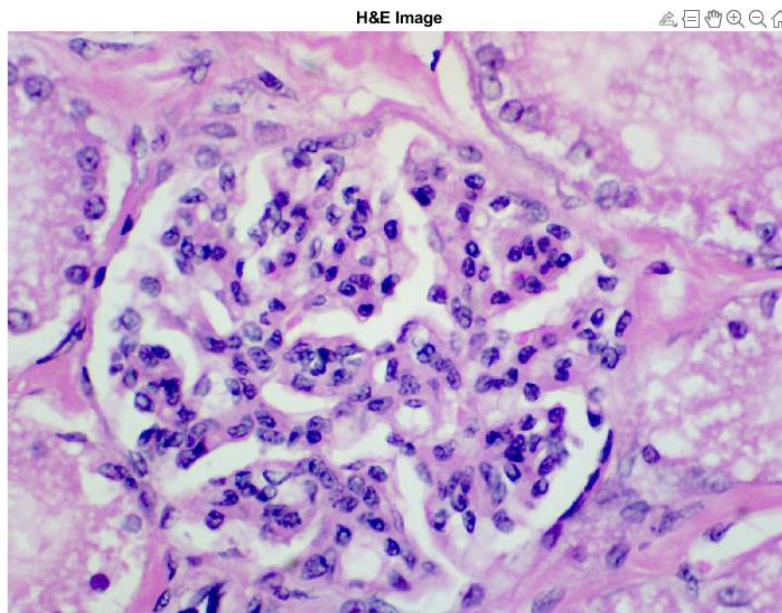
Objectives:

Clustering is a way to separate groups of objects. K-means clustering treats each object as having a location in space. It finds partitions such that objects within each cluster are as close to each other as possible, and as far from objects in other clusters as possible. You can use the `imsegkmeans` function to separate image pixels by value into clusters within a color space. This example performs k-means clustering of an image in the RGB and L*a*b color spaces to show how using different color spaces can improve segmentation results.

Step 1: Read Image

Read in `hestain.png`, which is an image of tissue stained with hemotoxylin and eosin (H&E). This staining method helps pathologists distinguish between tissue types that are stained blue-purple and pink.

```
he = imread("C:\Users\Robotic.lab\Downloads\img.jpeg");  
figure;  
imshow(he);  
title("H&E Image");
```

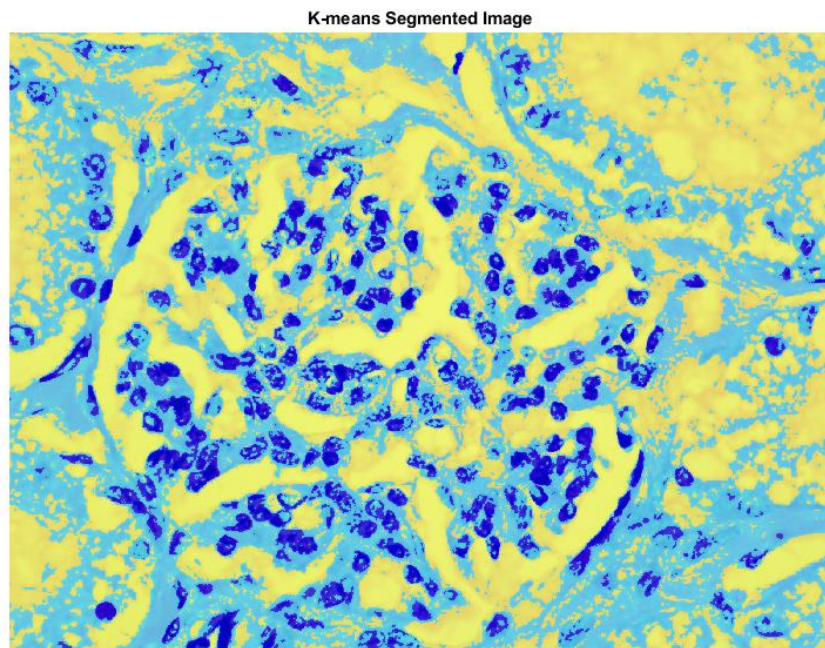


Step 2: Classify colors in RGB Color Space Using K-means Clustering

Segment the image into three regions using K-means clustering in the RGB color space. For each pixel in the input image, the `imsegkmeans` function returns a label corresponding to a cluster.

Display the label image as an overlay on the original image. The label image incorrectly groups white, light blue-purple, and light pink regions together. Because the RGB color space combines brightness color information within each channel (red, green, blue), lighter versions of two different colors are closer together and more challenging to segment than darker versions of the same two colors.

```
numcolors = 3;  
L = imsegkmeans(he, numcolors);  
B = labeloverlay(he, L);  
figure;  
imshow(B);  
title("K-means Segmented Image");
```



Step 3: Convert image from RGB Color Space to L*a*b Color Space

The L*a*b color space separates image luminosity and color. This makes it easier to segment regions by color, independent of lightness. The color space is also more consistent with human visual perception of the distinct white, blue-purple, and pink regions in the image.

The L*a*b color space is derived from the CIE XYZ tristimulus values. The L*a*b space consists of the luminosity layer L*, the chromaticity layer a* that indicates where a color falls along the red-green axis, and the chromaticity layer b* that indicates where a color falls along the blue-yellow axis. All of the color information is in the a* and b* layers.

Convert the image to the L*a*b color space by using the `rgb2lab` function.

```
lab_he = rgb2lab(he);
```

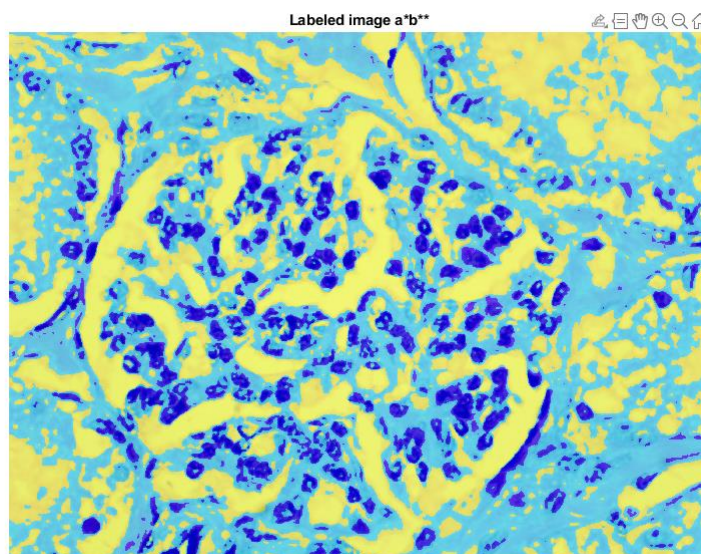
Step 4: Classify colors in a*b* Space using K-means Clustering

To segment the image using only color information, limit the image to the a* and b* values in `lab_he`. Convert the image to data type single for use with the `imsegkmeans` function to separate the image pixels into three clusters. Set the value of the `NumAttempts` argument to repeat clustering three times with different initial cluster centroid positions to avoid fitting to a local minimum.

```
ab=lab_he(:,:,2:3)
ab=im2single(ab)
pixel_labels=imsegkmeans(ab,numcolors,NumAttempts=3)
```

Display the label image as an overlay on the original image. The new label image more clearly separates the white, blue-purple, and pink stained tissue regions.

```
B2=labeloverlay(he,pixel_labels)
imshow(B2)
title("Labeled image a*b**")
```



Conclusion

The practical successfully demonstrated image segmentation using the K-means clustering algorithm in MATLAB, where the image was divided into different clusters based on color similarity, and improved segmentation was achieved by using the Lab* color space.